

Functional Requirement Document

Based on the provided Business Requirement Document (BRD), I will create a detailed Functional Requirement Document (FRD) adhering to the "Coding Requirements" section. Here are the functional requirements in bullet points:

FR-DLT-001: Load Customer and Order Data

Title: Load Customer and Order Data from CSV to Delta Tables

Description: The system must read customer and order data from CSV files and load it into delta tables.

Preconditions:

- CSV files are available at the specified locations (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata)
- Delta tables "customer" and "order" exist or can be created

Main Flow / Functional Steps:

- Read customer data from CSV file (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata) and load it into the "customer" delta table.
- Read order data from CSV file (/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata) and load it into the "order" delta table.

FR-DLT-002: Transform Order Data

Title: Add TotalAmount Column to Order Table

Description: The system must add a new column "TotalAmount" to the "order" table by multiplying "PricePerUnit" and "Qty" columns.

Preconditions:

- "order" delta table exists and has columns "PricePerUnit" and "Qty"

Main Flow / Functional Steps:

- Add a new column "TotalAmount" to the "order" table.
- Calculate the value of "TotalAmount" by multiplying "PricePerUnit" and "Qty" columns.

FR-DLT-003: Cleanse Customer and Order Data

Title: Remove Null and Duplicate Records from Customer and Order Tables

Description: The system must remove null and duplicate records from both "customer" and "order" tables.

Preconditions:

- "customer" and "order" delta tables exist

Main Flow / Functional Steps:

- Remove null records from "customer" and "order" tables.
- Remove duplicate records from "customer" and "order" tables.

FR-DLT-004: Create ordersummary Table

Title: Create ordersummary Table if Not Exists

Description: The system must create the "ordersummary" table if it does not exist in the specified catalog and schema.

Preconditions:

- Catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" exist

Main Flow / Functional Steps:

- Check if the "ordersummary" table exists in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Create the "ordersummary" table if it does not exist.

FR-DLT-005: Load ordersummary Table

Title: Load ordersummary Table with Joined Customer and Order Data

Description: The system must join "customer" and "order" tables on "CustId" and load the data into the "ordersummary" table as an SCD type 2 table.

Preconditions:

- "customer" and "order" delta tables exist and have been cleansed
- "ordersummary" table exists

Main Flow / Functional Steps:

- Join "customer" and "order" tables on "CustId".
- Load the joined data into the "ordersummary" table as an SCD type 2 table.

FR-DLT-006: Update ordersummary Table on Customer Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" table when there is a change in the "customer" table.

Preconditions:

- "customer" and "ordersummary" tables exist

Main Flow / Functional Steps:

- Detect changes in the "customer" table.
- Update the "ordersummary" table by making old records inactive and new records active.
- Update StartDate and EndDate accordingly to maintain history.

FR-DLT-007: Create customeragggregatespend Table

Title: Create customeragggregatespend Table if Not Exists

Description: The system must create the "customeragggregatespend" table if it does not exist in the specified catalog and schema.

Preconditions:

- Catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" exist

Main Flow / Functional Steps:

- Check if the "customeragggregatespend" table exists in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Create the "customeragggregatespend" table if it does not exist.

FR-DLT-008: Load customeragggregatespend Table

Title: Load customeragggregatespend Table with Aggregated Data

Description: The system must aggregate "TotalAmount" from "ordersummary" table and load the data into the "customeragggregatespend" table.

Preconditions:

- "ordersummary" table exists and has data
- "customeragggregatespend" table exists

Main Flow / Functional Steps:

- Aggregate "TotalAmount" from "ordersummary" table by "Name" and "Date".
- Load the aggregated data into the "customeragggregatespend" table.

FR-DLT-009: Implement using Delta Live Tables

Title: Implement Data Processing using Delta Live Tables

Description: The system must implement the data processing pipeline using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented

Main Flow / Functional Steps:

- Implement the data processing pipeline using Delta Live Tables, covering all previous requirements.